

Qatar-10b

R-0143

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250717 · created 2026-07-16T07:59:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460873.859 +/- 0.013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.051
Transit depth (Rp/Rs)^2	2.97 +/- 1.76 [%]
Orbital Inclination (inc)	89.32 +/- 2.63
Airmass coefficient 1 (a1)	0.382 +/- 0.063
Airmass coefficient 2 (a2)	0.62 +/- 0.48
Scatter in the residuals of the lightcurve fit is	17.06 %
Best Comparison Star	#1 - [475, 362]
Optimal Aperture	1.75
Optimal Annulus	5.87
Transit Duration (day)	0.1229 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.051 vs 0.1265 (deviation 0.89σ)
Duration fit vs published	0.1229 d vs 0.1154 d (deviation 1.13σ)
Mid-time O–C	15.1 min
Depth SNR	3.4
Residual scatter	17.06 %

Light curve

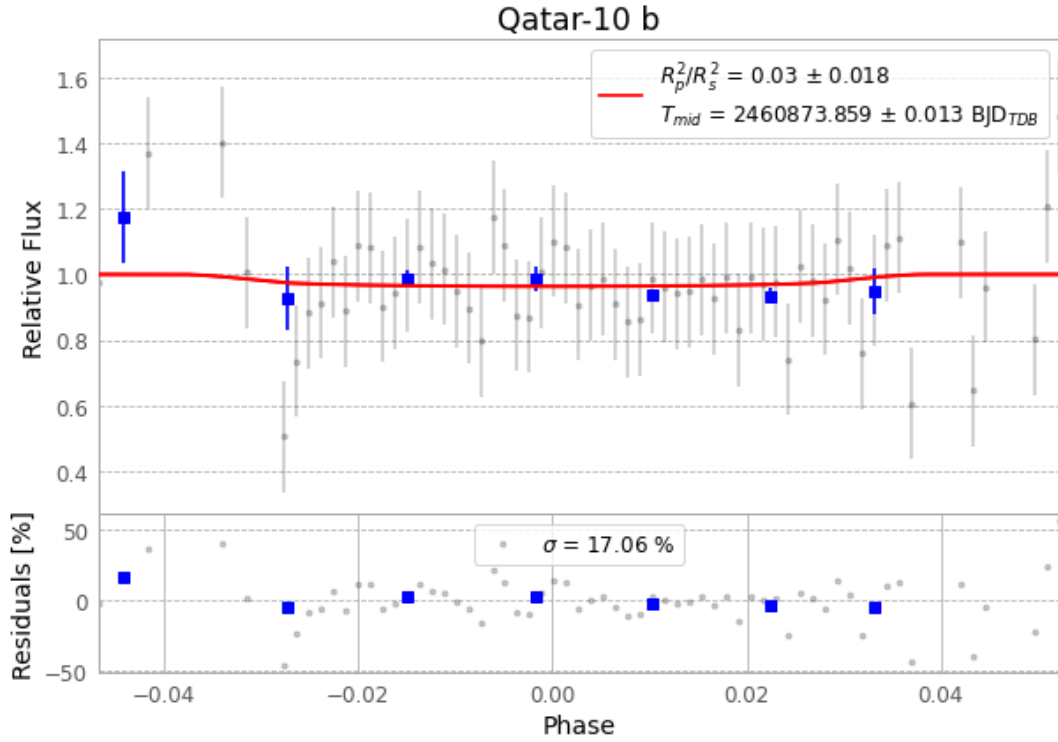


Figure 1 — FinalLightCurve_Qatar-10 b_2025-07-16.png (SHA-256 f914c29c0f726d7a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [365, 287], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8bed43af4652b839...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

79 FITS frames (52 MB), Qatar-10250717064515.FITS → Qatar-10250717103915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250717.log (377e36b68e44...), FinalLightCurve_Qatar-10 b_2025-07-16.png (f914c29c0f72...), FinalLightCurve_Qatar-10 b_2025-07-16.csv (298a4f532db8...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 07:59Z.